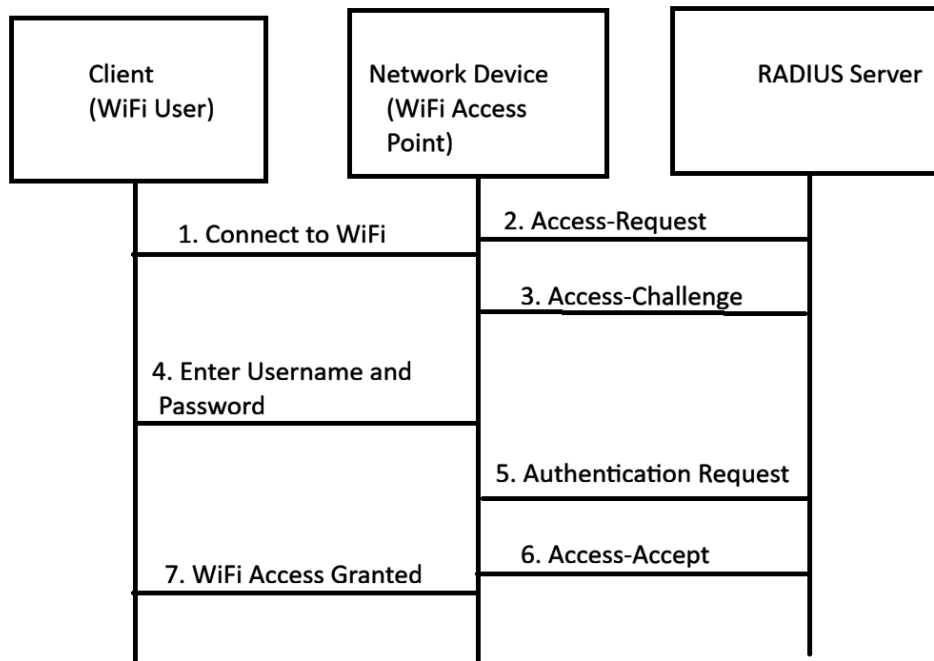


D.RADIUS AND TACACS

1.Create a comparison table listing at least 4 key features of RADIUS and TACACS+ side by side, highlighting which protocol supports each feature.

Feature	RADIUS	TACACS+
Authentication	Supports authentication	Supports authentication
Authorization	Limited authorization support	Full authorization support
Accounting	Supports accounting	Supports accounting
Transport Protocol	Uses UDP	Uses TCP
Encryption	Encrypts the user password	Encrypts the entire packet body
AAA Separation	Authentication and authorization are combined	Authentication, authorization and accounting are separated
Port Number	UDP 1812 for authentication and UDP 1813 for accounting	TCP 49
Common Usage	Used for network access, Wi-Fi and VPN authentication	Mainly used for network device administration

2.Draw a step-by-step flow diagram showing the RADIUS authentication process for a user logging in to a public WiFi hotspot, labeling each communication step between the client, network device, and RADIUS server.



3. List 3 real-world use cases where TACACS+ is preferred over RADIUS, and briefly explain why TACACS+ is better suited for each scenario.

No.	Use Case	Why TACACS+ is Better Suited
1	Network Device Administration	TACACS+ provides separate authentication and authorization, allowing administrators to have different levels of access to routers, switches and firewalls.
2	Large Enterprise Networks	TACACS+ provides detailed control over administrator permissions and can record the commands executed by users on network devices.
3	Network Operations and Security Teams	TACACS+ allows organizations to control which commands each administrator can execute. This helps prevent unauthorized configuration changes and improves accountability.

4. Given a scenario where a user tries to log into a network device but fails authentication, describe how RADIUS and TACACS+ would handle the logging and response differently. Hint: Focus on how each protocol manages accounting and command authorization.

Point	RADIUS	TACACS+
Authentication Failure	RADIUS server rejects the login request and sends an Access-Reject response.	TACACS+ server rejects the authentication request and sends a failure response.
Logging	RADIUS can record authentication and accounting information, such as login sessions.	TACACS+ provides detailed logging of administrator authentication and activities.
Accounting	RADIUS mainly records session information such as start, stop and duration.	TACACS+ can provide detailed accounting of administrator activities and commands.
Command Authorization	RADIUS has limited support for controlling individual commands.	TACACS+ can authorize individual commands, allowing or denying specific commands.

5. Use ChatGPT or another AI tool to generate a sample RADIUS configuration snippet for a Cisco router, then review the output and list any changes you would make to adapt it for a college campus WiFi setup.

